Innate immunity · exam pearls one-pager

EAACI/UEMS Knowledge Exam, 19 Oct 2026 · **!** = exam trap · Source tags: A = Abbas 11e, J = Janeway 8e, GL = WAO/EAACI HAE guideline 2017 (page numbers; hover for fact ID) · amber tag = volatile.

M1 Foundation

- Innate receptors: germline-encoded, nonclonal, <100 types; memory limited. A64,4
- Phases: immediate innate 0–4 h, early induced 4–96 h, adaptive >96 h. J37
- Paneth cells make α-defensins. LL-37 (cathelicidin) neutralizes LPS. A80
- TNF, IL-1, IL-6 → fever, leukocytosis, acute-phase proteins; CRP can rise 1000-fold. A96,97,88
- Epithelial IL-25, IL-33, TSLP drive ILC2s, Th2 cells, mast cells; IL-33 is an alarmin. A94

! **Trap:** lysozyme works better on Gram-positive bacteria (LPS covers Gram-negative peptidoglycan). J45

! **Trap:** anti-TNF gave no benefit in sepsis trials and can reactivate TB. A97 J109

M2 Cells

- Neutrophils: most abundant WBC, arrive first; azurophilic granules hold MPO and defensins. A14,94
- Most tissue macrophages are fetal-derived and self-renew (Janeway: from monocytes). A15 J76
- pDC (TLR7, TLR9) = major type I IFN source; cDC1 cross-presents to CD8+ T cells. A23,22,21-23
- NK cells: CD56+ CD3−; low MHC-I ("missing self") removes KIR/NKG2A inhibition; CD16 → ADCC. A82,85,84
- NK deficiency (GATA2, CD16, CHS) → severe herpesvirus and papillomavirus infections. A488
- C3a, C5a degranulate mast cells; MRGPRX2 → IgE-independent drug reactions. A470
- ILC2 (GATA3; IL-5, IL-13) responds to IL-25, IL-33, TSLP. A81

! **Trap:** tryptase, not histamine, marks mast cell activation (all mast cells, no other cell). A471

! **Trap:** eosinophil FcεRI is low and lacks a signalling chain; IL-5 is the key cytokine. A473

M3 PRRs & signalling

- Surface TLR1/2/4/5/6 see wall products; endosomal TLR3/7/8/9 see nucleic acids. A69
- LPS → TLR4 (+MD2, CD14); flagellin → TLR5; dsRNA → TLR3; CpG DNA → TLR9. A67,69
- MyD88/IRAK4 deficiency → early invasive pneumococcal disease, no fever ("cold"). A489 J535
- TLR3, UNC93B, TRIF defects → herpes simplex encephalitis. A489
- NLRP3–ASC–caspase-1 cleaves pro-IL-1β/IL-18; gasdermin D pores → pyroptosis. A72,74
- NLRP3 gain of function → CAPS (FCAS: cold-induced; Muckle-Wells: deafness); IL-1 blockade works. A75,74 J537

! **Trap:** TLR3 is the only TLR without MyD88; TLR4 uses MyD88 and TRIF. A69

! **Trap:** weak NOD2 variants → Crohn's; NOD2 gain of function → Blau syndrome. A72

M4 Phagocytosis & defects

- Recruitment: selectin rolling (sialyl Lewis X) → chemokine integrin activation → ICAM-1/VCAM-1 arrest → transmigration. A49,45
- CXCL8 (CXCR1/2) → neutrophils; CCL2 (CCR2) → monocytes. A50
- CGD: about 2/3 X-linked gp91; catalase+ *Staphylococcus*; *Aspergillus* = leading cause of death; granulomas. A486,487
- LAD-1 (CD18): leukocytosis, no pus, poor wound healing. LAD-2: no sialyl Lewis X, Bombay group. LAD-3 (KINDLIN-3): bleeding. A488
- CHS (LYST): partial albinism, giant lysosomes, NK defect, HLH. A488
- MSMD (IL-12p40, IL-12Rβ1, IFN-γR, STAT1, NEMO): mycobacteria, *Salmonella*, BCG spread. A498

! **Trap:** NADPH oxidase makes superoxide; MPO makes hypohalous acid. MPO deficiency is CGD-like but milder. A96 J536

M5 Complement

- C3 cleavage is central. C3 convertases: C4b2a (classical, lectin), C3bBb (alternative). A301,305 J54
- IgM fixes complement better than IgG; IgG1, IgG3 best; IgG4 none. A305
- Factor D circulates active; properdin stabilizes C3bBb. J61 A302
- C5a: most potent anaphylatoxin; degranulates mast cells without IgE. A314,470
- CR2 (CD21) binds C3d, is the EBV receptor and helps B-cell activation. A308,310,314

! **Trap:** factor H regulates only the alternative pathway; C4BP regulates C4b. A311,310

! **Trap:** CD59 blocks C9 at C5b-8; DAF dissociates the C3 convertases. A312,310

M6 Deficiencies & HAE

- C1q/C2/C4 → SLE (>50%); C2 deficiency is the most common. A315
- C3 or factor I deficiency → pyogenic infections. A315
- C5–C9, properdin (X-linked), factor D → *Neisseria*. A315 J532
- aHUS: factor H, MCP. PNH: somatic PIG-A → no CD59, DAF. A315,312
- HAE-1: low C1-INH function and protein. HAE-2: protein normal/high, function low. C4 usually low. GL5,6
- AAE-C1-INH: later onset, lymphoma/MGUS, C1q low in 75%. GL7
- ACE inhibitors: avoid in HAE; ACEI-AE has normal C1-INH. GL14,6

! **Trap:** C3 and CH50 are normal in HAE; C4 is unreliable before 12 months. GL6-7,11

! **Trap:** HAE-1/2 does not respond to antihistamines, steroids or adrenaline; wheals do not exclude HAE. GL6,7

2017 guideline (volatile — check current): treat attacks early with C1-INH, ecallantide or icatibant; no tranexamic acid or androgens on demand. LTP: C1-INH first line, androgens second. GL7,7-8,8,9-10,10